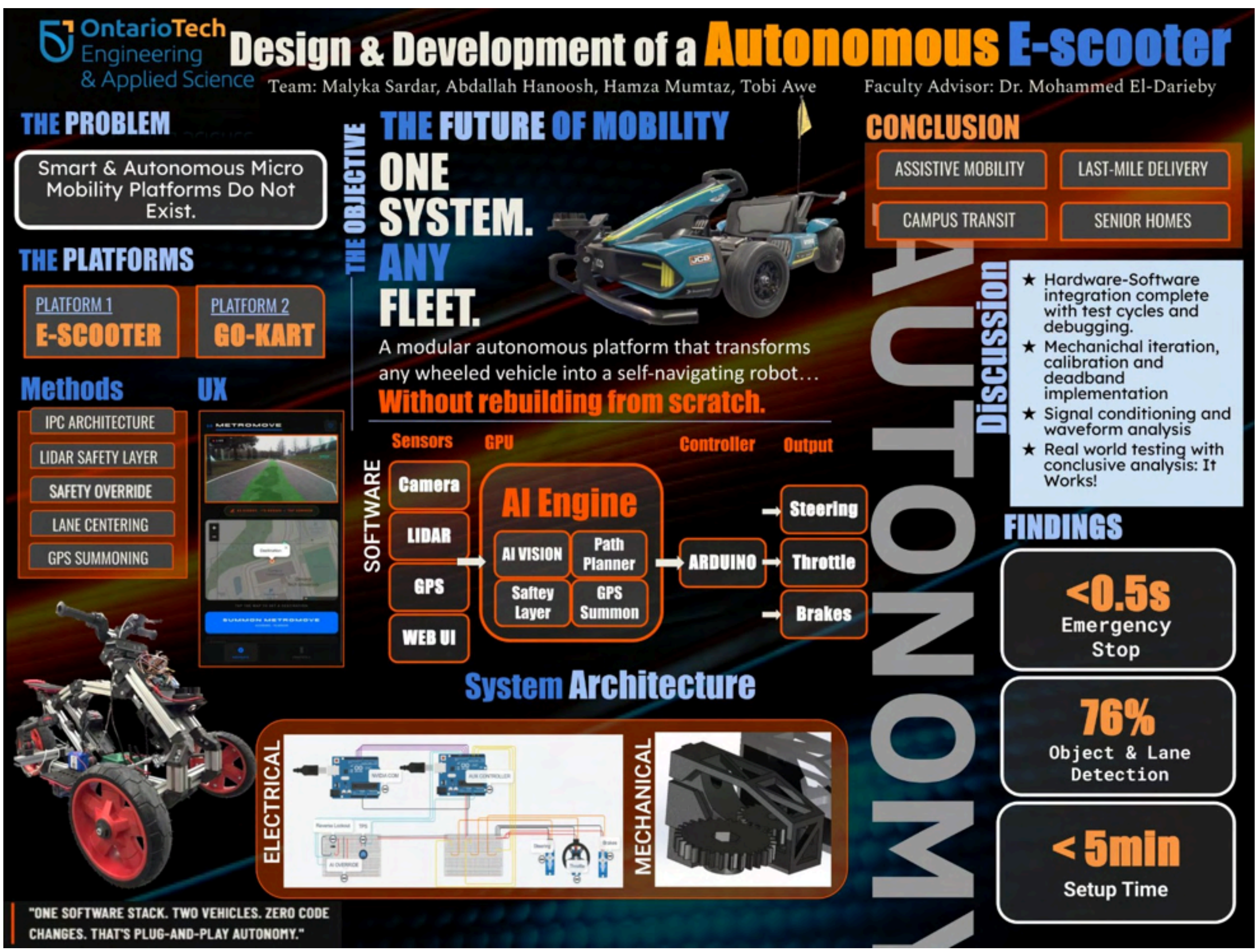
Capstone Project — Autonomous E-Scooter

Embedded & Autonomous Systems Integration



Project Snapshot

- Team: 4 (2 Electrical, 2 Software)
- Focus: Embedded control systems, CAN communication, and distributed system integration
- Scope: Autonomous electric scooter platform

My Ownership

- **Architectural Lead:** Served as the primary hardware-software interface lead, owning embedded control architecture, CAN bus network implementation, and physical subsystem integration.
- **Bus Design & Protocol:** Engineered the distributed CAN communication architecture, defining message identifiers, signal scaling, bus speed, and fault-handling routines to ensure deterministic data exchange under load.
- **Subsystem Integration:** Integrated heterogeneous sensors and actuators (LiDAR, GPS, cameras, servo steering, electronic braking), enforcing strict signal integrity, grounding practices, and real-time responsiveness.
- **Validation & HIL Testing:** Performed systematic Hardware-in-the-Loop (HIL) testing, signal conditioning, and iterative oscilloscope waveform analysis to isolate and resolve cross-domain bugs.

Key Technical Challenges

- **Deterministic Communication:** Maintaining reliable, low-latency serial data transmission across multiple distributed nodes while managing bus load and fault isolation.
- **Signal Integrity & Latency:** Integrating complex analog/digital sensors while preserving signal fidelity and loop stability in a high-noise electrical environment.
- **System-Level Troubleshooting:** Resolving complex electro-mechanical feedback issues spanning firmware timing, wiring impedance, and real-time control constraints.

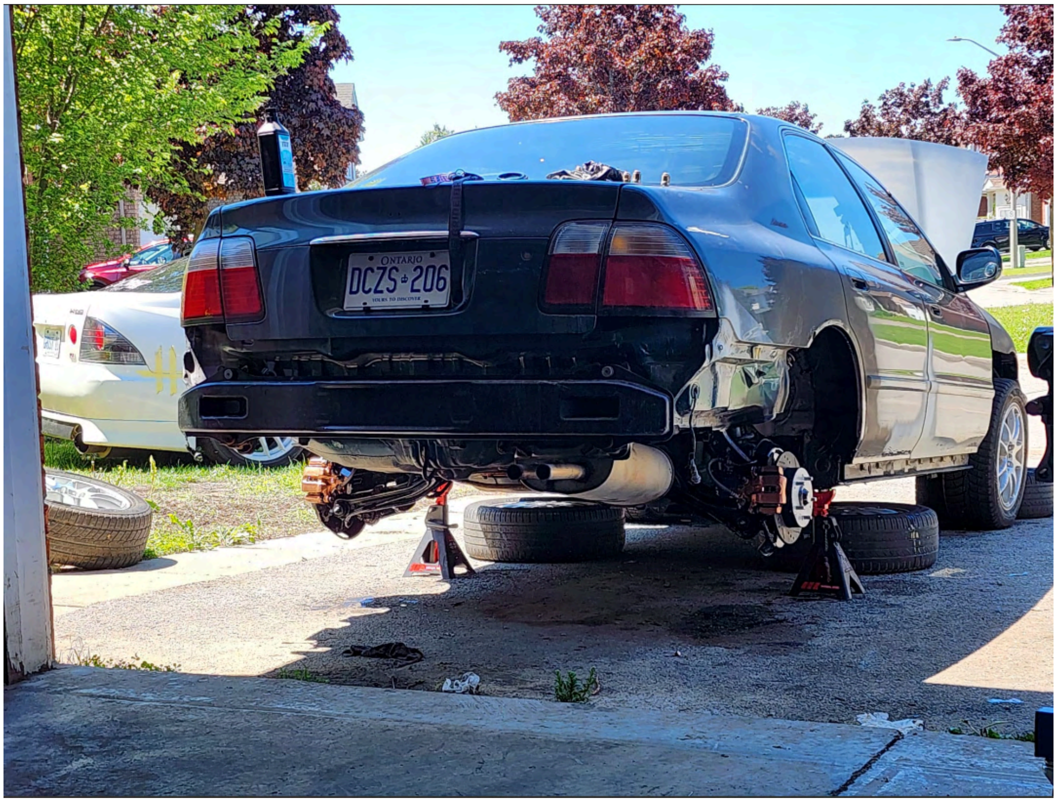
Engineering Summary

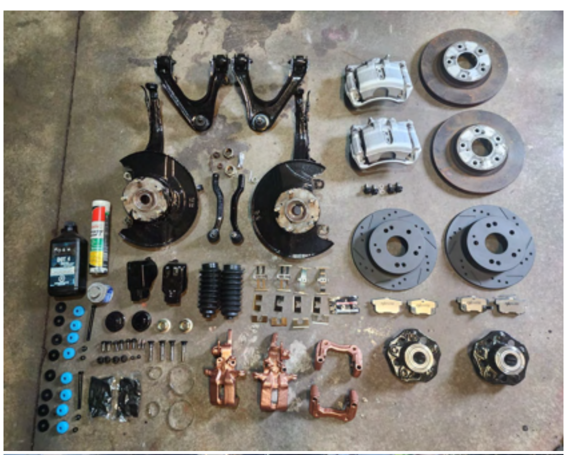
This project required approaching the system as a whole rather than as isolated components. Many integration issues only became visible once hardware, firmware, and software were operating together, requiring iterative debugging and close coordination with the software team. My role focused on identifying system-level failure points, validating communication and control behaviour, and ensuring that embedded hardware constraints were clearly understood and reflected in the overall system design. This methodical approach delivered a reliable platform capable of sub-0.5-second emergency stops and 76% object/lane detection accuracy.

Tools & Technologies

CAN bus, embedded microcontrollers, sensors & actuators, sensor integration, oscilloscope waveform analysis, HIL testing, system validation.

Honda Accord — (JTCC COMPLIANT)
Vehicle Systems Engineering & Embedded Controls





Project Snapshot

- **Platform:** 1996 Honda Accord (CD5)
- **Focus:** Focus on high-performance engine management, custom sensor packaging, and vehicle-level electrical architecture.
- **Scope:** Long-term, self-directed vehicle development project inspired by JTCC touring car platforms

My Ownership

- **Full-Lifecycle Ownership:** Responsible for the complete system design, integration, and validation of the vehicle's mechanical, electrical, and embedded subsystems.
- **Engine Management & Calibration:** Designed custom standalone engine management configurations, mapping fuel/ignition tables and implementing embedded control logic.
- **Wiring & Harness Architecture:** Engineered and fabricated custom wiring layouts, strictly routing power, ground planes, and shielded sensor lines to protect signal integrity.
- **Sensor Packaging & Integration:** Integrated custom sensor suites for high-resolution engine position detection, pressure monitoring, and vehicle telemetry.
- **Validation & Troubleshooting:** Performed systematic hardware testing, signal conditioning, and iterative refinement to isolate cross-domain faults.

Key Technical Challenges

- **Trigger Mechanism Design:** Designing reliable custom camshaft and crankshaft trigger configurations to achieve precise phase detection without signal loss at high RPM.
- **Signal Integrity & EMI Mitigation:** Integrating aftermarket sensors while maintaining timing accuracy and mitigating voltage drops and electrical noise in a harsh under-hood environment.
- **Mechanical-Electrical Co-Design:** Coordinating complex mechanical modifications (engine internals, balance shaft delete) with embedded control and monitoring requirements.

Engineering Summary

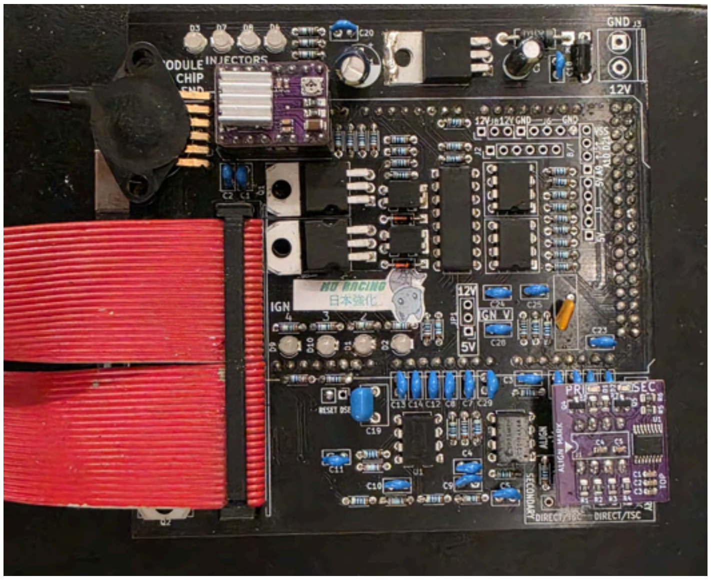
This project is centered on treating a production vehicle as a tightly coupled mechatronic system rather than a collection of independent modifications. Many challenges only emerge when mechanical changes, harsh under-hood conditions, and embedded control logic interact in real operating conditions. My work focused on designing sensing and control solutions that are electrically robust, mechanically compliant, and safe for high-performance operation, validating behavior through structured testing and iteration.

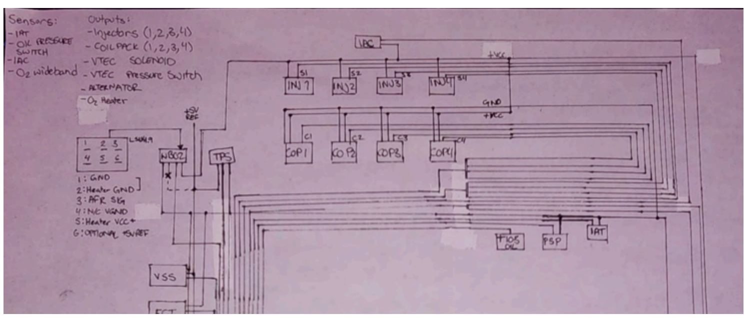
Tools & Technologies

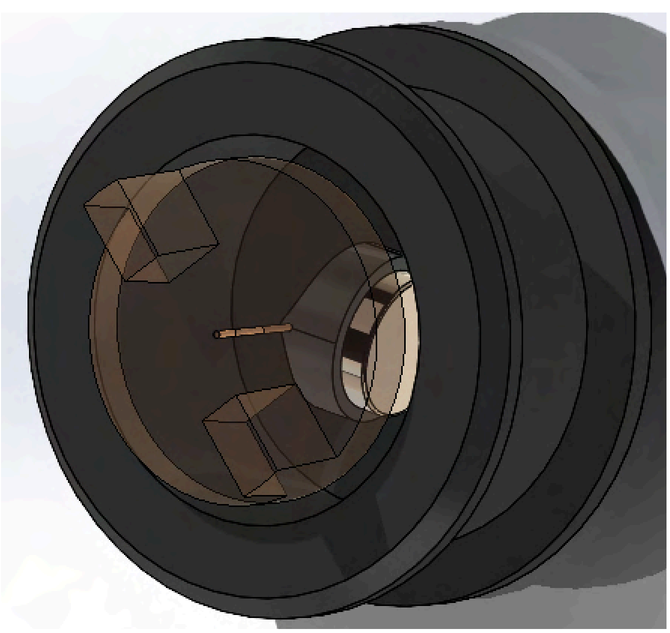
Standalone ECU configuration, custom wiring harness fabrication, engine position sensors, CAN-based telemetry, oscilloscope diagnostics, mechanical design and fabrication, system testing & validation.

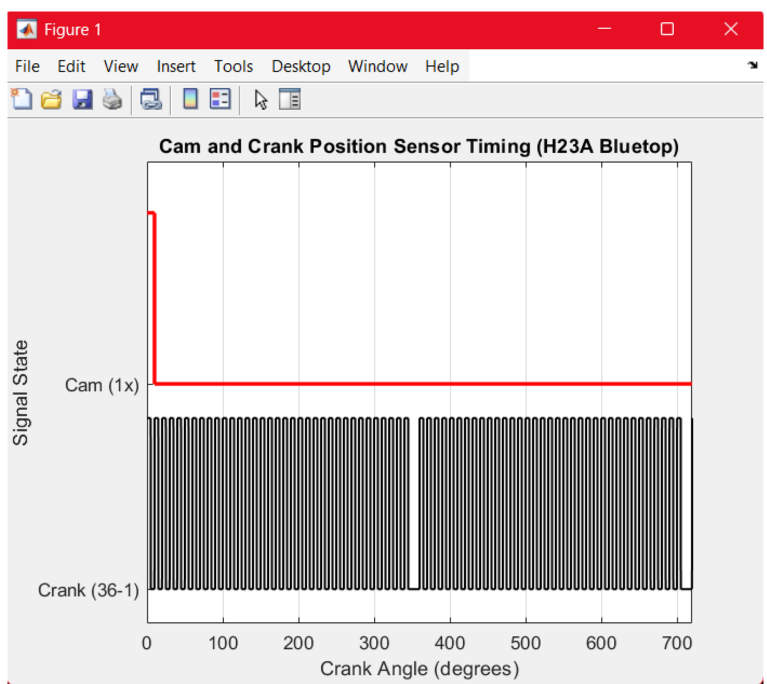
Custom ECU & Engine Management Systems

Embedded Systems, Sensors & Timing-Critical Control









Project Snapshot

Focus: Embedded engine management, timing-critical sensing, hardware integration

Scope: Custom engine management and sensing system for performance automotive applications

My Ownership

- **System-Level Ownership:** Responsible for the complete design, implementation, and validation of a custom engine management and instrumentation system, emphasizing timing accuracy and real-world reliability.
- **Sensing & Reluctor Design:** Designed custom camshaft and crankshaft position sensing configurations, optimizing reluctor wheel geometry to achieve required timing resolution and sensor compatibility.
- **ECU Configuration & Wiring:** Configured standalone ECU parameters, wiring architecture, and signal conditioning pathways to interface engine instrumentation (boost pressure, RPM, phase).
- **Signal Conditioning & Noise Mitigation:** Engineered robust signal pathways and filtering to protect high-frequency pulse trains and analog inputs from severe automotive electromagnetic noise.
- **Calibration & Validation:** Performed sensor calibration, telemetry data logging, and systematic testing under real operating conditions to isolate cross-domain faults.

Key Technical Challenges

Designing a reliable engine management system required solving timing-critical challenges where small errors in sensing or signal quality could directly impact system behavior.

- **Timing-Critical Resolution:** Achieving accurate, repeatable, and low-latency cam/crank position sensing where minor signal degradation could disrupt ignition or fueling timing.
- **Electrical Noise & Interference:** Managing severe under-hood electrical noise while preserving absolute signal integrity across multi-channel sensor arrays.
- **System Synchronization:** Synchronizing complex sensor inputs to feed stable, real-time RPM, phase, and load metrics into the core control logic.

Engineering Summary

This project required treating engine management as a tightly coupled embedded system rather than a collection of independent sensors. Many issues only became visible once mechanical design, sensor behavior, wiring, and embedded configuration interacted under real operating conditions.

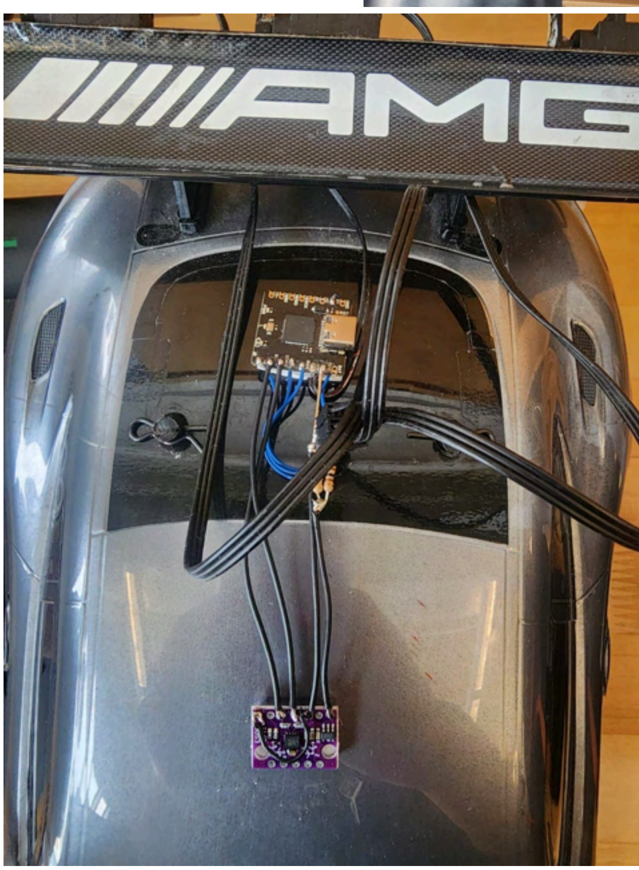
My work focused on identifying timing and signal integrity constraints, validating sensor behavior through testing and calibration, and refining the system to achieve stable and predictable operation. The resulting system demonstrated reliable engine position detection, consistent instrumentation readings, and embedded behavior suitable for integration into a larger vehicle control architecture.

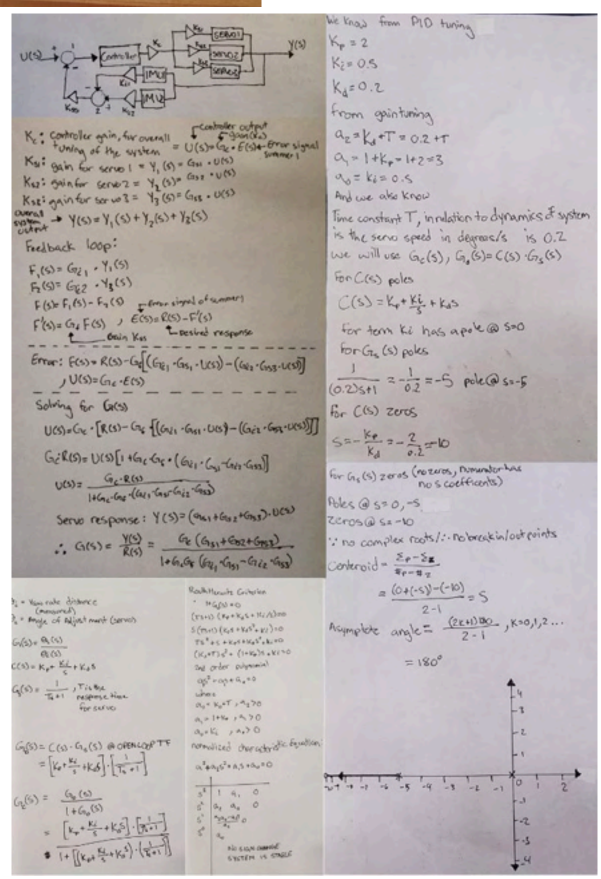
Tools & Technologies

Embedded ECU, custom reluctor wheel design, engine position sensors, wiring harness design, signal conditioning, oscilloscope diagnostics, telemetry & data logging, system validation.

Active Aerodynamic Control System (1/10 Scale Vehicle)

Closed-Loop Control & Sensor-Driven Actuation





Project Snapshot

Focus: Closed-loop control systems, sensor integration, real-time actuation

Scope: Actively controlled aerodynamic system developed for a 1/10 scale vehicle to study dynamic response during braking and cornering

My Ownership

- **Control Systems Ownership:** Led the complete design, implementation, and validation of an active aerodynamic control architecture, bridging IMU sensor fusion to real-time servo actuation.
- **Firmware Logic & PD Control:** Developed proportional-derivative (PD) closed-loop control algorithms in firmware to dynamically modulate aerodynamic surfaces based on live vehicle states.
- **Sensor Integration & Fusion:** Integrated on-board Inertial Measurement Units (IMUs) to capture real-time pitch, roll, yaw, and deceleration loads.
- **Mechanical Packaging:** Designed mechanical linkages and optimized servo motor placement to handle high-frequency actuation cycles under physical aerodynamic loads.
- **System Tuning & Validation:** Performed systematic calibration, filtering optimization, and iterative testing to refine control behavior under dynamic driving conditions.

Key Technical Challenges

- **Noise Mitigation & Filtering:** Filtering high-frequency road vibration and structural noise from IMU telemetry to prevent erratic actuator jitter.
- **Actuator Dynamics & Limits:** Balancing response speed against physical mechanical torque limits to maintain stable real-time aerodynamic control.
- **Multi-Domain Debugging:** Isolating cross-domain faults spanning sensor signal conditioning, firmware execution logic, and physical mechanical actuation.

Engineering Summary

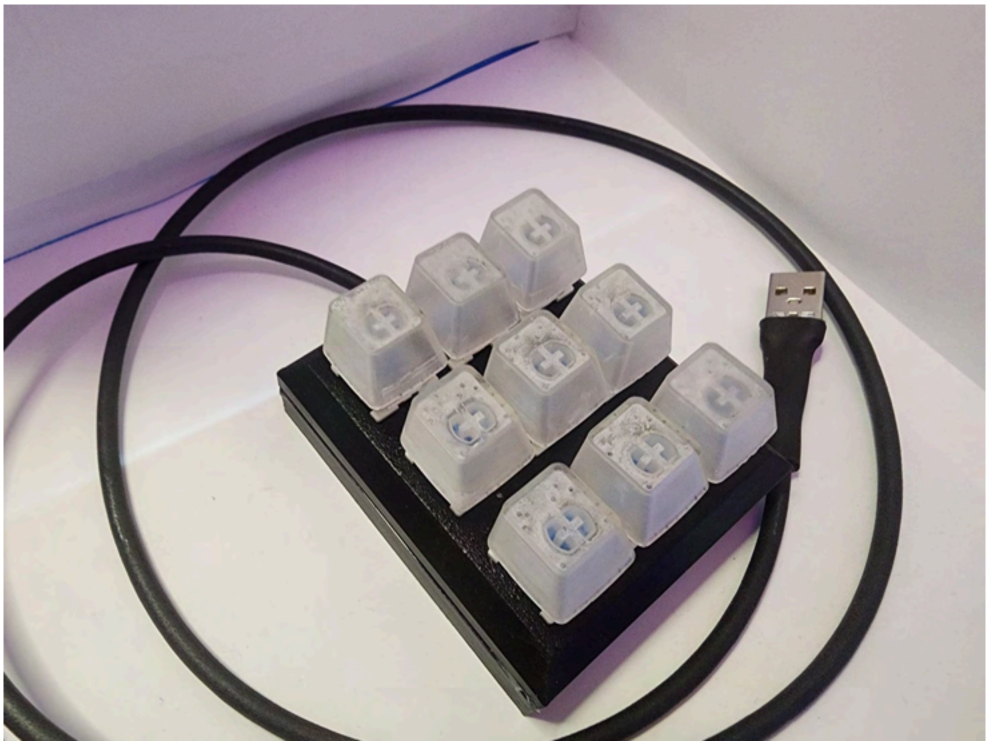
This project bridged classical control theory with physical mechatronics, requiring tight coupling between sensing, computation, and actuation. Many challenges only emerged once the system operated dynamically under real-world forces, requiring empirical tuning of filter coefficients and control gains. My work focused on ensuring sensor reliability, smooth control responses, and precise mechanical execution, yielding a platform capable of predictive aerodynamic shifts during dynamic driving maneuvers.

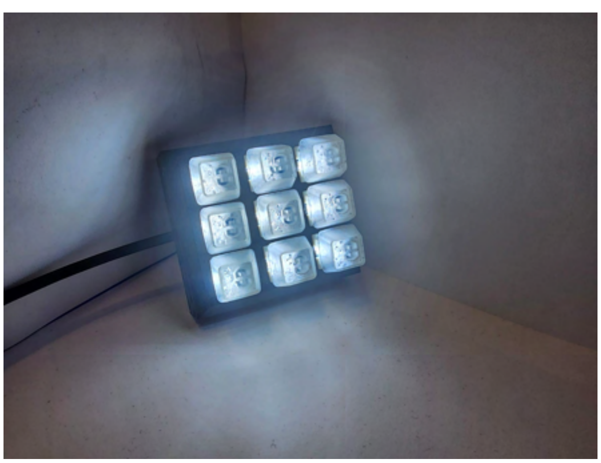
Tools & Technologies

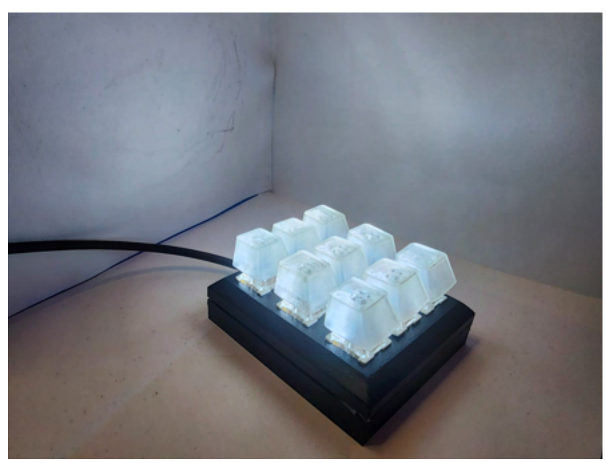
Embedded microcontroller, inertial sensors (IMUs), servo actuators, closed-loop control logic, mechanical integration, system testing & validation

Adaptive / Assistive Keyboard

Human-Centered Design & Accessible Hardware Development







Project Snapshot

Focus: Human-centered engineering, requirements-driven development, and accessible hardware design

Scope: Custom adaptive keyboard developed to support users with physical or cognitive accessibility needs under strict academic and functional criteria.

My Ownership

- **Requirements Capture & Translation:** Translated complex ergonomic and accessibility criteria into rigorous functional hardware specifications, governing key travel, layout geometry, and actuation limits.
- **Architecture & Circuit Design:** Engineered the underlying matrix switch wiring and microcontroller firmware to support reliable input scanning, debouncing, and custom macro execution.
- **Mechanical Packaging:** Manufactured the physical enclosure and internal mounting plates in CAD, ensuring structural rigidity, component protection, and clean cable management.
- **Prototyping & Assembly:** Handled full-lifecycle component selection, layout, and physical prototyping of the adaptive input device.
- **Documentation & Verification:** Authored comprehensive technical documentation and design reports to communicate trade-offs, engineering decisions, and verification results to faculty evaluators.

Key Technical Challenges

- **Translating Human Factors:** Converting qualitative accessibility constraints into quantitative engineering parameters without compromising technical feasibility.
- **Reliability Under Customization:** Designing a modular layout and robust matrix architecture that accommodates alternative interaction profiles while maintaining electrical stability.
- **Latency & Tactile Response:** Minimizing switch debounce latency in firmware while preserving a crisp, repeatable tactile response for the user.

Engineering Summary

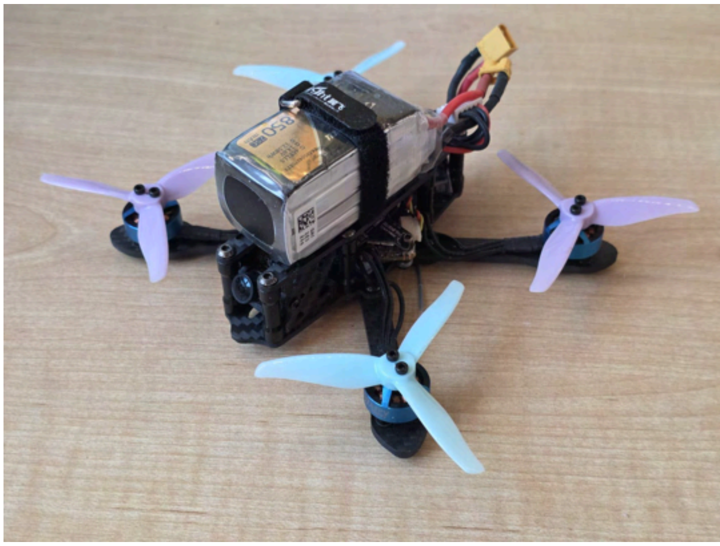
This project reinforced the critical intersection of structured requirements management, formal documentation, and human-centered design. Success required ensuring that complex technical choices remained entirely subordinate to the end-user’s physical interaction needs. By following a rigorous, documented design process from requirements capture to prototype validation, the project met strict academic standards and demonstrated how structured engineering practices apply directly to accessibility-focused hardware.

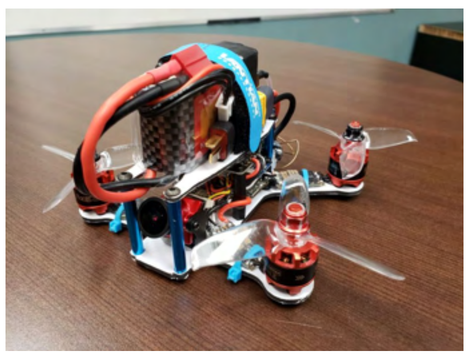
Tools & Technologies

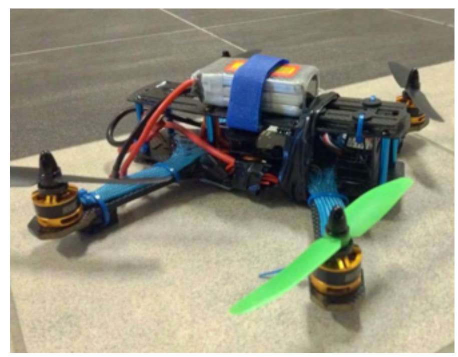
Microcontroller programming, matrix switch wiring, CAD enclosure design, rapid prototyping, technical documentation, design verification.

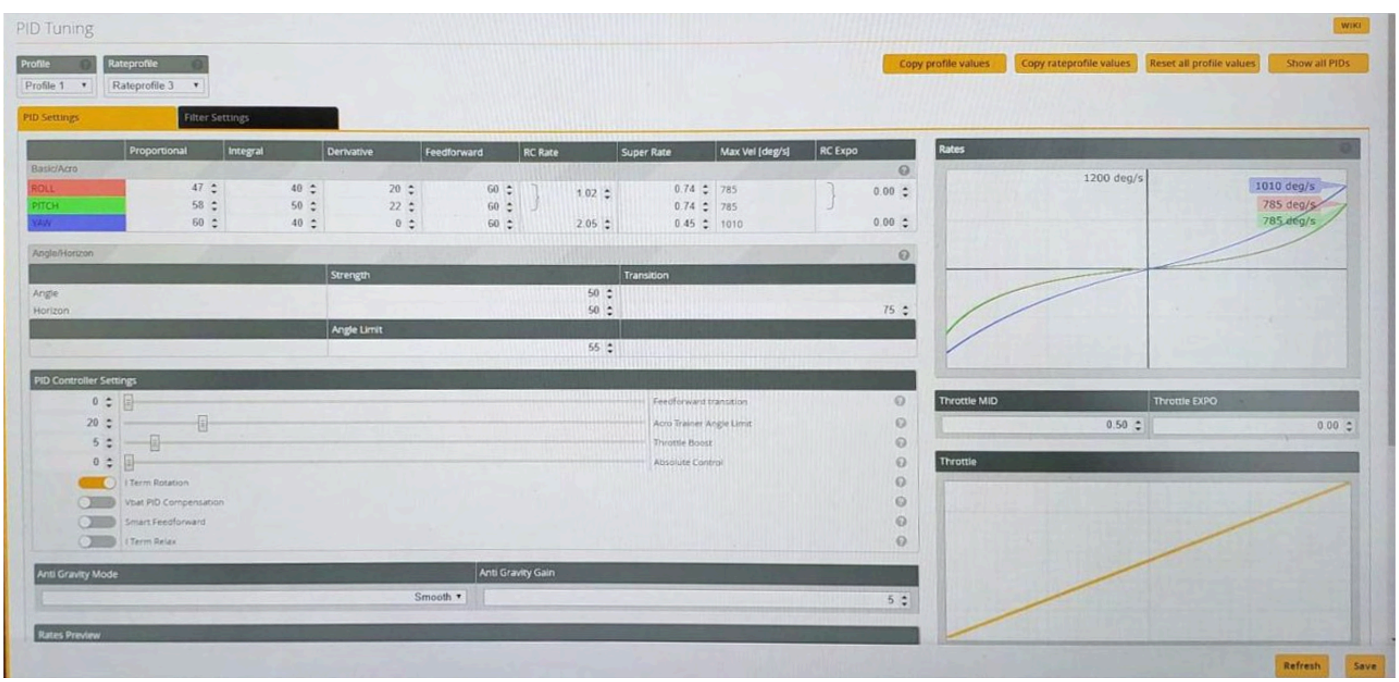
High-Performance UAV Platform & Flight Control Systems

Custom Multi-Rotor Design, Embedded Control & Environmental Validation









Project Snapshot

Focus: Flight control systems, embedded mechatronics, and rigorous hardware resilience

Scope: Iterative design and development of four custom FPV multirotor UAV platforms to analyze closed-loop stability, power distribution constraints, and real-world system dynamics under extreme operating conditions.

My Ownership

- **System-Level Iteration:** Led the full-lifecycle design, fabrication, and testing of four distinct multirotor platforms, advancing from custom Atmel-based microcontroller architectures to high-performance ARM Cortex-F4 and F7 processing cores.
- **Control Loop Tuning:** Configured flight control firmware, utilizing Blackbox telemetry logging to analyze IMU/accelerometer noise frequencies, tune dynamic PID coefficients, and eliminate high-RPM mechanical resonance.
- **Ruggedized Design & Optimization:** Engineered a high-performance, sub-250g platform optimized for strict weight-to-power constraints, capable of reaching 114 km/h.
- **Environmental Hardening:** Application of industrial-grade silicone conformal coating and rigorous hardware insulation strategies to protect sensitive onboard electronics from severe environmental moisture and electrical breakdown.

Key Technical Challenges

- **Extreme Disturbance Rejection:** Maintaining control-loop stability, sensor fidelity, and hardware integrity under high-G maneuvers and aggressive aerodynamic loads.
- **Environmental Validation:** Ensuring electronic survival and continuous communication/control functionality during severe adverse weather operations.
- **Resonance & Noise Mitigation:** Isolating high-frequency mechanical vibration and electrical noise to prevent sensor saturation and actuator jitter.

Engineering Summary

This project represents a progressive, multi-year deep dive into practical control systems and mechatronic design. By iterating through four physical platforms, from early custom-soldered Atmel/MPU6050 setups to advanced F7-based systems, I focused on bridging theoretical control loops with harsh, real-world physical dynamics. The culmination of this work was a lightweight, high-speed (114 km/h) sub-250g UAV rigorously validated through adverse environmental testing, including maintaining stable flight and successful flight inside a severe thunderstorm.

Tools & Technologies

ARM Cortex-F7/F4 flight controllers, custom firmware configuration, Blackbox telemetry & data logging, PID control tuning, electronic noise filtering, conformal coating, rapid prototyping, system testing & validation.